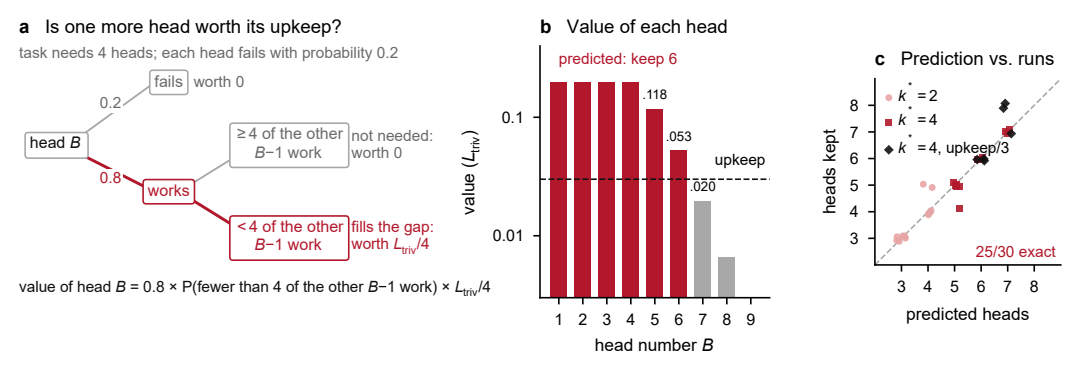
a Is one more head worth its upkeep?
task needs 4 heads; each head fails with probability 0.2
head B
0.2
fails
worth 0
0.8
works
≥ 4 of the other B−1 work
not needed: worth 0
< 4 of the other B−1 work
fills the gap: worth L_triv/4
value of head B = 0.8 × P(fewer than 4 of the other B−1 work) × L_triv/4
b Value of each head
predicted: keep 6
.118
.053
.020
upkeep
value (L_triv)
0.1
0.01
1 2 3 4 5 6 7 8 9
head number B
c Prediction vs. runs
k* = 2
k* = 4
k* = 4, upkeep/3
heads kept
8 7 6 5 4 3
3 4 5 6 7 8
predicted heads
25/30 exact